

Analog IC Design

Design of Rail-to-Rail Operational Transconductance Amplifier (OTA)

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I. INTRODUCTION

The objective of this project is to design and simulate a Rail-to-Rail Operational Transconductance Amplifier (OTA) using the SCL 180 nm CMOS technology node. The aim is to ensure that the design specifications of the OTA are met. The design specifications are mentioned in the Table 1.

Parameter	Requirement
DC Gain	≥ 70 dB
Unity Gain Bandwidth	≥ 40 MHz
Phase Margin	$60^\circ - 85^\circ$
Input Common Mode Range (ICMR)	200 mV to $(V_{supply} - 0.2)$ V
Output Common Mode Range (OCMR)	200 mV to $(V_{supply} - 0.2)$ V
Common Mode Rejection Ratio (CMRR)	≥ 90 dB
Load Capacitance	≥ 5 pF
Power consumption	≤ 1 mW
Supply Voltage	≤ 1.8 V

TABLE I: Design Specifications

All specifications must be satisfied across process, voltage and temperature (PVT) variations, considering a 10% supply variation and a temperature range from -40°C to 125°C .

II. RAIL-TO-RAIL OTA

A Rail-to-Rail OTA is a special type of OTA that can handle input and/or output signals that span the entire supply voltage range, that is, from the negative rail (V_{SS}) to the positive rail (V_{DD}). The reduction in power supply voltages in the new and upcoming CMOS technologies leads to a reduction in the dynamic range of the circuits. However, a low signal swing is not good and it is important to maximize the signal swing. It will not only ensure a high dynamic range but will also ensure a good signal-to-noise ratio (SNR). A high dynamic range is possible only with rail-to-rail amplifiers. A rail-to-rail OTA has several applications such as in low-power analog filters, data converters and sensors.

III. RAIL-TO-RAIL OTA ARCHITECTURE

The complete circuit diagram of a rail-to-rail OTA implemented in this project is shown in figure 1.

It consists of 5 main blocks:

- 1) Input stage
- 2) Summing circuit

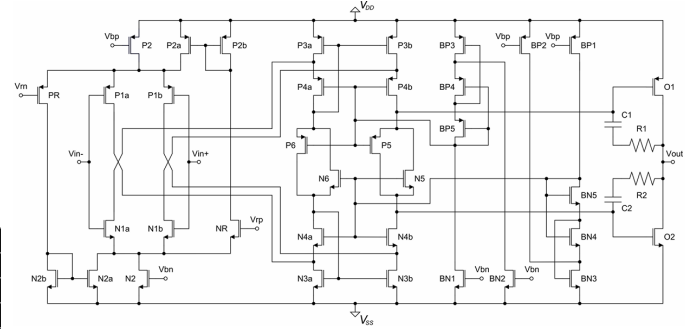


Fig. 1: Complete Rail-to-Rail OTA circuit

- 3) Class-AB control block
- 4) Bias block
- 5) Output stage

A. Input Stage

The input stage of the rail-to-rail OTA consists of NMOS and PMOS differential pairs connected in parallel as shown in figure 2. The positive and negative terminal input signals are given to both NMOS and PMOS. The NMOS input pair is active when the input voltage is between $V_{DSat} + V_{GSn}$ and V_{DD} and the PMOS input pair is active only when the input voltage is between V_{SS} and $V_{DD} - (V_{DSat} + V_{GSp})$. Therefore, as it can be seen from figure 3, atleast one of the two transistors (NMOS or PMOS) is always active for any common mode voltage given at the input.

However, the issue here is that the transconductance will vary with the common mode voltage, which we do not want. In the middle, both NMOS and PMOS will be active and at the higher or lower voltages, only one of them will be active. Hence, the transconductance g_m will vary by a factor of almost 2. The calculation of the total transconductance $g_{m_{tot}}$ of the input stage is as shown below:

$$g_{m_{tot}} = g_{m_n} + g_{m_p}$$

$$g_{m_n} = \sqrt{2\mu_n C_{ox} \frac{W_n}{L_n} I_{Dn}} \quad ; \quad g_{m_p} = \sqrt{2\mu_p C_{ox} \frac{W_p}{L_p} I_{Dp}}$$

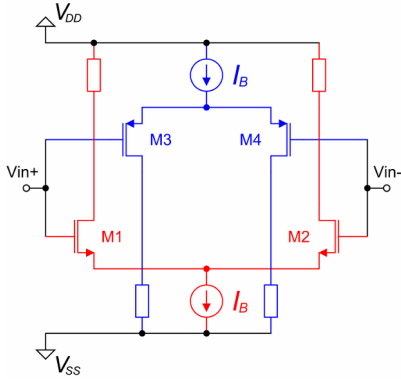


Fig. 2: Input stage circuit diagram

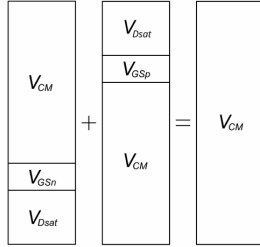


Fig. 3: Active voltage regions of NMOS, PMOS and the combined input stage

Thus, if the W/L ratios of NMOS and PMOS are adjusted based on the ratios of the product of their mobility and gate capacitance, their drain current I_{Dn} and I_{Dp} can be made equal. Therefore, the total transconductance in terms of the currents will be:

$$gm_{tot} = C(\sqrt{I_{Dn}} + \sqrt{I_{Dp}})$$

where C is a constant. In order to maintain a constant gm_{tot} , when one of the current is zero, the other needs to be multiplied by 4. This can be done by ensuring that three times the drain current is added only when needed. Therefore, the circuit shown in figure 4 can be used for this.

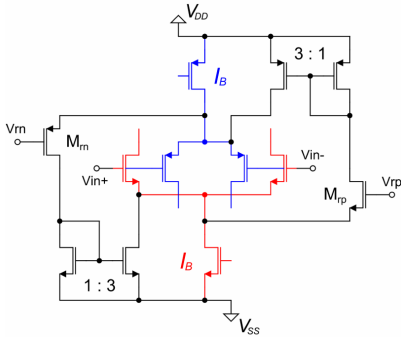


Fig. 4: Constant gm input stage

In this circuit, the transistors M_{rn} and M_{rp} act as switches. When a low common mode voltage is applied, M_{rn} is off and

M_{rp} is conducting and when a high common mode voltage is applied, M_{rn} is conducting and M_{rp} is off. Therefore, in both cases, it is ensured that the total current is $4I_B$. In this way, a constant transconductance input stage can be made.

Figure 5 shows the plots of the transconductances of NMOS pair, PMOS pair and the overall variation of transconductance of the input stage with common mode voltage.

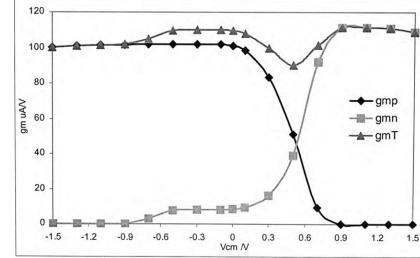


Fig. 5: Variation of gm of the input stage

B. Summing Circuit

The input stage has 4 output currents, one each at the drains of the four transistors used in the two differential pairs to which the two input signals are given. Hence, these currents need to be added now and this is done using the summing circuit shown in figure 6. It is made using a self-biased folded-cascode structure. The advantage of this kind of a configuration is that no external bias network is required for it.

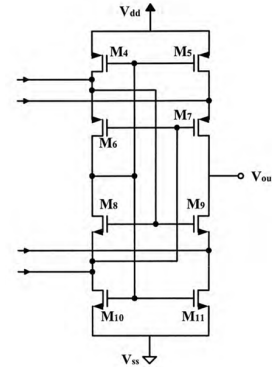


Fig. 6: Summing stage circuit diagram

C. Class-AB Control Block

The class-AB control block consists of the transistors P5, N5 and P6, N6. It is placed in between the summing circuit as shown in figure 7. This block plays a critical role in biasing the push-pull output stage of the rail-to-rail OTA. The PMOS and NMOS output transistors receive just enough gate bias, from this control block, to stay slightly on at zero input. It enables the output stage to operate in such a way that it provides high output current when needed while consuming low current.

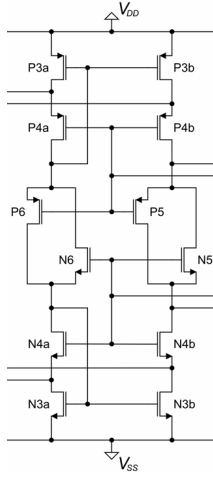


Fig. 7: Class-AB control circuit diagram

D. Bias Block

The Bias block consists of the transistors BP1, BP2, BP3, BP4, BP5, BN1, BN2, BN3, BN4 and BN5 as shown in figure 8. The role of this block is to generate the bias voltages for the cascodes and the floating current sources. The necessary bias DC currents are generated and then copied to the corresponding transistors.

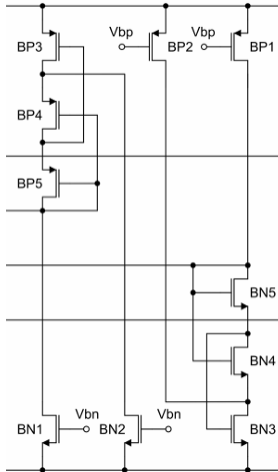


Fig. 8: Bias block circuit diagram

E. Output Stage

The output stage is the last stage of the rail-to-rail OTA design and it is biased using the class-AB control stage described earlier. It consists of two common-source connected transistors, O1 and O2 and these two transistors are directly driven by two currents I_{in1} and I_{in2} respectively as shown in the circuit figure 9. C1, C2 are frequency compensation capacitors, and R1 and R2 are zero-nulling resistors.

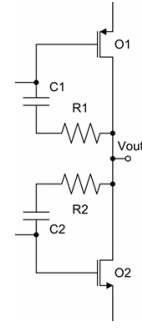


Fig. 9: Output stage circuit diagram

IV. SMALL SIGNAL ANALYSIS OF RAIL-TO-RAIL OTA

A simplified small signal model of half of the amplifier in open loop configuration is shown in Figure 9. In this small signal model, gm_1 is the transconductance of the first stage, gm_2 is the is the transconductance of the second stage, R_1 is the output resistance of the first stage, R_2 is the output resistance of the second stage, C_c is the compensation capacitance and C_L is the load capacitance.

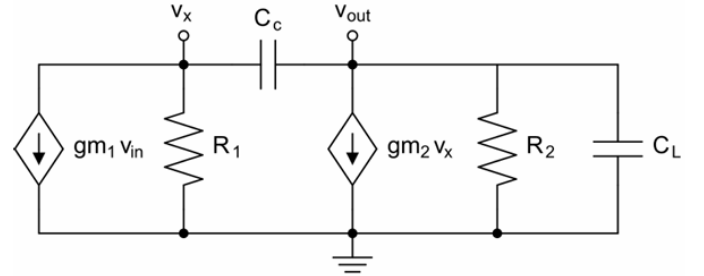


Fig. 10: Small Signal Analysis of half rail-to-rail OTA

By applying KCL at the x node and the output node, the following equations can be obtained:

$$gm_1 v_{in} + \frac{v_x}{R_1} + (v_x - v_{out})sC_c = 0$$

$$gm_2 v_x + \frac{v_{out}}{R_2} + sC_L v_{out} + (v_{out} - v_x)sC_c = 0$$

By solving these two equations, we will obtain the below transfer function for the circuit:

$$\frac{v_{out}(s)}{v_{in}(s)} = \frac{gm_1 R_1 R_2 (gm_2 - sC_c)}{C_c C_L R_1 R_2 s^2 + (C_L R_2 + C_c (R_1 + R_2 + gm_2 R_1 R_2))s + 1}$$

The above transfer equation can be written in the form of zeros and poles as shown below:

$$\frac{v_{out}}{v_{in}} = \frac{A_{OL} \left(1 + \frac{s}{z_1}\right)}{\left(1 + \frac{s}{p_1}\right) \left(1 + \frac{s}{p_2}\right)}$$

The denominator of the transfer function can be expanded as form:

$$\left(1 + \frac{s}{p_1}\right) \left(1 + \frac{s}{p_2}\right) = 1 + s \left(\frac{1}{p_1} + \frac{1}{p_2}\right) + \frac{s^2}{p_1 p_2}$$

As the non-dominant pole p_2 is generally much larger the dominant pole p_1 in order to ensure stability of the circuit, the following simplification can be made in the transfer function:

$$\frac{v_{out}}{v_{in}} = \frac{A_{OL} \left(1 + \frac{s}{z_1}\right)}{1 + \frac{s}{p_1} + \frac{s^2}{p_1 p_2}}$$

Now, from the transfer function obtained for the circuit, the values of the terms in the generalised transfer function can be identified. The open-loop gain of the OTA is obtained:

$$A_{OL} = A_1 A_2 \approx gm_1 gm_2 R_{out1} R_{out2}$$

To find the zero of the OTA, the numerator can be equated to 0 as shown below:

$$gm_2 - sC_c = 0$$

$$z_1 = \frac{gm_2}{C_c}$$

The dominant pole p_1 of the OTA can be found out by equating the term in the denominator that is multiplied by s to 0 as follows:

$$C_L R_2 + C_c (R_1 + R_2 + gm_2 R_1 R_2) = \frac{1}{p_1}$$

$$p_1 = \frac{1}{C_c R_1 + C_c R_2 + C_L R_2 + C_L gm_2 R_1 R_2} \approx \frac{1}{C_c gm_2 R_1 R_2}$$

The non-dominant pole p_2 of the OTA can be obtained by equating the term in the denominator that is multiplied by s^2 to 0 as follows:

$$C_c C_L R_1 R_2 = \frac{1}{p_1 p_2}$$

$$p_2 = \frac{1}{C_c C_L R_1 R_2 p_1} = \frac{gm_2}{C_L}$$

Also, the unity gain bandwidth of the amplifier can be obtained by multiplying the open-loop gain of the amplifier by its bandwidth. As the dominant pole is the generally limits the bandwidth of the amplifier, the unity gain bandwidth GBW can be obtained as follows:

$$GBW = A_{OL} p_1 = \frac{gm_1}{C_c}$$

V. CALCULATIONS

The values of various parameters for NMOS and PMOS for the SCL 180nm technology are:

$$u_n C_{ox} = 264 \mu A/V^2$$

$$u_p C_{ox} = 65 \mu A/V^2$$

$$r_{on} = 11 k\Omega$$

$$r_{op} = 8.4 k\Omega$$

As shown in part III A, the total transconductance of the first stage gm_1 can be given as follows:

$$gm_1 = gm_n + gm_p$$

$$gm_n = \sqrt{2\mu_n C_{ox} \frac{W_n}{L_n} I_{Dn}} \quad ; \quad gm_p = \sqrt{2\mu_p C_{ox} \frac{W_p}{L_p} I_{Dp}}$$

The total transconductance of the second stage can be given by the follows:

$$gm_2 = gm_{O1} + gm_{O2}$$

where gm_{O1} and gm_{O2} are the transconductances of the O1 and O2 transistors of the output stage. The requirement for the DC gain of the Rail-to-Rail OTA is 70 dB which is a gain of 3162.277. The DC gain of the OTA can be given by:

$$A_v = gm_1 gm_2 R_1 R_2$$

The output impedance R_1 can be given by $(R_{on} || R_{op})$ and that of the output stage can be given by $(R_{O1} || R_{O2})$. Therefore, the values of R_1 and R_2 would be:

$$R_1 = R_2 = 4.9 k\Omega$$

Therefore, the required value of the product $gm_1 gm_2$ is:

$$gm_1 gm_2 = \frac{A_v}{R_1 R_2}$$

$$gm_1 gm_2 = 131.69 \mu A^2/V^2$$

Therefore, I can keep the value of gm_1 to be around 30 mA/V and the value of gm_2 to be around 4.33 mA/V.

VI. RAIL-TO-RAIL OTA SIMULATION

The design of the rail-to-rail OTA as proposed in the earlier sections was implemented in Cadence Virtuoso as shown in figure 11.

In the circuit, IN+ is the non-inverting terminal, IN- is the inverting terminal and OUT is the output terminal of the OTA. The positive rail voltage V_{DD} is 1.8 V and the negative rail voltage V_{SS} is 0 V. The sizing of all the transistors in the circuit is given in table II.

Note that the names of the transistors, resistors and capacitors are based on the circuit schematic shown in figure 1.

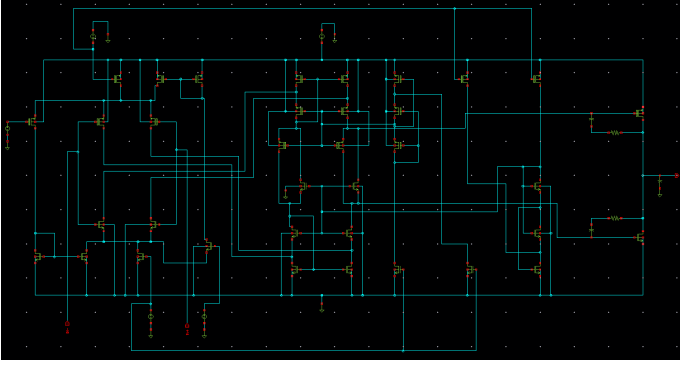


Fig. 11: Implementation of Rail-to-Rail OTA in Cadence Virtuoso

Components	W/L Sizing (in μm) or Value
P2, N2	20/0.36
P2b, N2b	10/0.36
BP1	3/0.36
BN3	30/0.36
BP2, BN1	0.36/0.36
BN2, BN4	40/0.36
P1a, P1b, N1a, N1b, PR, NR	100/0.36
P2a, N2a	60/0.36
P3a, P3b	56/0.36
N3a, N3b	24/0.36
P4a, P4b, BP4	20/0.36
N4a, N4b	10/0.36
P5, P6, N5, N6, BN5, BP3, BP5	100/0.36
C1, C2	100 pF
R1, R2	0 k Ω
O1, O2	5/0.36

TABLE II: Design Parameter Sizing and Values

VII. RESULTS AND OBSERVATIONS

A. DC Operating Point Analysis

From the figure 5, it can be seen the maximum input stage transconductance is obtained for a common mode voltage close to 0.9 V and the least input stage transconductance is obtained for a common mode voltage close to 0.6 V.

On performing the DC operating point analysis for a common mode voltage of 0.6V, the total current flowing through the input stage NMOS and PMOS are 1.4853 mA and 2.512 mA. Therefore, the values of transconductances gm_n and gm_p can be given by the formula:

$$gm_n = \sqrt{2\mu_n C_{ox} \frac{W_n}{L_n} I_{Dn}} \quad ; \quad gm_p = \sqrt{2\mu_p C_{ox} \frac{W_p}{L_p} I_{Dp}}$$

$$gm_n = 14.756 \text{ mA/V}$$

$$gm_p = 9.521 \text{ mA/V}$$

Therefore, the value of gm_1 will be 24.277 mA/V.

Also, the total current flowing through the input stage NMOS and PMOS are 37.942 μA and 37.932 μA . So, the values of transconductances gm_{O1} and gm_{O2} from the same formula will be:

$$gm_{O1} = 2.359 \text{ mA/V}$$

$$gm_{O2} = 1.170 \text{ mA/V}$$

Therefore, the values of gm_2 will be 3.529 mA/V. Thus, the product $gm_1 gm_2$ is $85.67533 \mu\text{A}^2/\text{V}^2$.

Now, on performing the DC operating point analysis for a common mode voltage of 0.9V, the total current flowing through the input stage NMOS and PMOS are 3.8541 mA and 1.918 mA. Therefore, the values of transconductances gm_n and gm_p can be obtained from the formula:

$$gm_n = 23.776 \text{ mA/V}$$

$$gm_p = 8.329 \text{ mA/V}$$

Therefore, the value of gm_1 will be 32.105 mA/V.

Also, the total current flowing through the input stage NMOS and PMOS are 117.489 μA and 117.482 μA . So, the values of transconductances gm_{O1} and gm_{O2} from the same formula will be:

$$gm_{O1} = 4.152 \text{ mA/V}$$

$$gm_{O2} = 2.059 \text{ mA/V}$$

Therefore, the values of gm_2 will be 6.211 mA/V. Thus, the product $gm_1 gm_2$ is $199.4 \mu\text{A}^2/\text{V}^2$.

So, the maximum value of transconductance of the input stage is 32.105 mA/V and the minimum value is 24.277 mA/V. Therefore, the variation in transconductance is:

$$\frac{\Delta gm}{gm} = \frac{32.105 - 24.277}{32.105} = 0.2438$$

Therefore, the variation in transconductance is close to 24.38%.

B. Open Loop Transient Analysis

Figure 12 shows the input signal at the non-inverting terminal (in red), input signal at the inverting terminal (in yellow) and the output signal (in green) of the Rail-to-Rail OTA when a sine wave of amplitude 10 μV and frequency 1 kHz is given at the input with a common mode voltage of 1.1 V. The output signal voltage has a peak-to-peak amplitude of 90 mV.

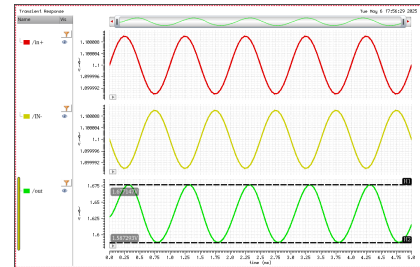


Fig. 12: Open-Loop Transient Analysis for Common Mode Voltage = 1.1 V

Figure 13 shows the input signal at the non-inverting terminal (in red), input signal at the inverting terminal (in yellow) and the output signal (in green) of the Rail-to-Rail OTA when a sine wave of amplitude 50 μ V and frequency 100 kHz is given at the input with a common mode voltage of 0.9 V. The output signal voltage has a peak-to-peak amplitude of 15.6 mV.

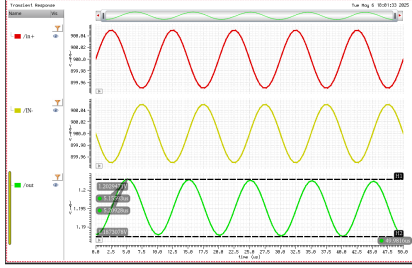


Fig. 13: Open-Loop Transient Analysis for Common Mode Voltage = 0.9 V

C. AC Analysis

Figure 14 shows the plot for the open-loop amplitude gain in dB for a frequency range of 1 Hz to 1 GHz when AC analysis is done with a common mode voltage of 0.8 V.

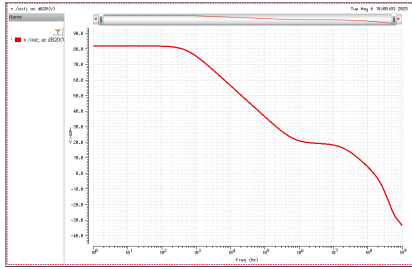


Fig. 14: AC Analysis for Common Mode Voltage = 0.8

Figure 15 shows the plot for the open-loop amplitude gain in dB for a frequency range of 1 Hz to 1 GHz when AC analysis is done with a common mode voltage of 1.2 V.

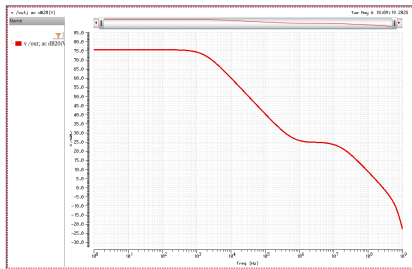


Fig. 15: AC Analysis for Common Mode Voltage = 1.2

Table III summarizes the results obtained after the AC analysis of the Rail-to-Rail OTA such as DC gain and -3 dB frequency.

V_{CM} (in V)	DC Gain (in dB)	-3dB Frequency (in MHz)
0.4	70.1	113.743
0.5	69.6655	90.1138
0.6	60.024	36.24
0.7	71.0065	77.0796
0.8	81.904	188.743
0.9	101.3794	330.575
1	73.76	411.227
1.1	73.437	407.584
1.2	75.565	360.217
1.3	70.981	219.409

TABLE III: AC Analysis Results

D. Stability Analysis

Figure 16 shows the plot for the magnitude (in blue) and phase (in green) obtained on performing stability analysis with a common mode voltage of 0.5 V.

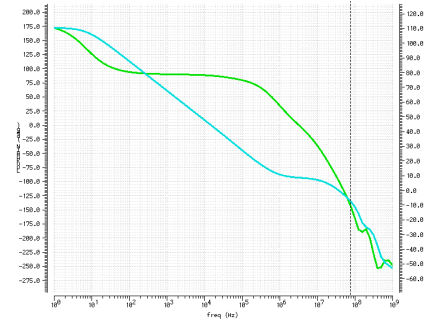


Fig. 16: Stability Analysis for Common Mode Voltage = 0.5 V

Figure 17 shows the plot for the magnitude (in violet) and phase (in blue) obtained on performing stability analysis with a common mode voltage of 1 V.

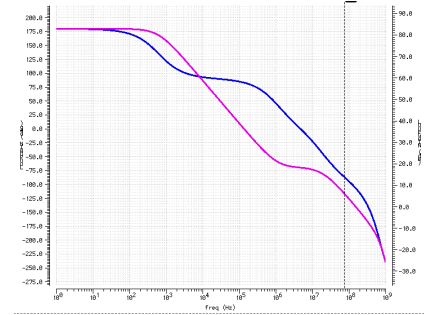


Fig. 17: Stability Analysis for Common Mode Voltage = 1 V

The table IV summarises the results obtained after performing stability analysis, for various common mode voltages, such as DC gain, Unity Gain Bandwidth (UGB) and Phase margin (PM).

E. Power Consumption

Figure 18 shows the plot for current flowing through the power supply V_{DD} when the common mode voltage is 0.7 V. The value of V_{DD} is 1.8 V and the current value is 5.714

V_{CM} (in V)	DC Gain (in dB)	UGB (in MHz)	PM (in degrees)
0.4	111.2417	38.1867	98.63
0.5	110.382	37.3024	80.44
0.6	105.1423	2.621	184.9
0.7	102.666	28.335	91.73
0.8	103.9095	78.4353	67.9
0.9	96.963	132.217	68.4
1	82.9063	154.591	73.22
1.1	79.442	149.088	76.176
1.2	73.975	127.452	80.13
1.3	60.346	76.737	92.61

TABLE IV: Stability Analysis Results

mA. Therefore, the total power consumption is 10.2852 mW.

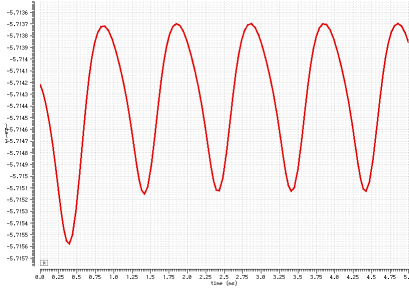


Fig. 18: Total Supply Current for Common Mode Voltage = 0.7 V

Figure 18 shows the plot for current flowing through the power supply V_{DD} when the common mode voltage is 0.5 V. The value of V_{DD} is 1.8 V and the current value is 2.249 mA. Therefore, the total power consumption is 4.0482 mW.

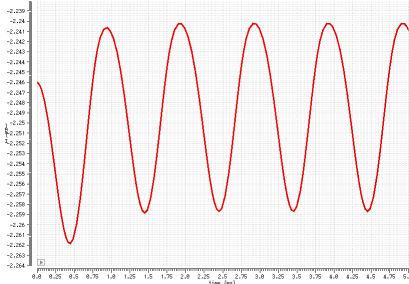


Fig. 19: Total Supply Current for Common Mode Voltage = 0.5 V

F. CMRR Analysis

The table V shows the results obtained, such as common mode gain and CMRR (Common Mode Rejection Ratio), on performing CMRR analysis of the Rail-to-Rail OTA for various common mode voltages.

G. PSRR Analysis

The table VI shows the results obtained on performing PSRR (Power Source Rejection Ratio) analysis of the Rail-to-Rail OTA for various common mode voltages.

V_{CM} (in V)	Common Mode Gain (in dB)	CMRR (in dB)
0.4	-19.364	89.464
0.5	-10.147	79.812
0.6	-5.34	65.364
0.7	-1.10	72.10
0.8	5.787	76.117
0.9	31.66	69.713
1	10.899	62.87
1.1	5.932	67.50
1.2	682.85m	74.883
1.3	-6.592	82.157
1.4	-18.567	89.014

TABLE V: CMRR Analysis Results

V_{CM} (in V)	PSRR (in dB)
0.4	-52.352
0.5	-42.9392
0.6	-33.815
0.7	-25.034
0.8	-12.412
0.9	15.821
1	-18.648
1.1	-13.006
1.2	-8.876
1.3	-4.9311
1.4	-1.4817

TABLE VI: PSRR Analysis Results

H. Noise Analysis

Figure 20 shows the plot for the input-referred noise voltage on carrying out noise analysis on the Rail-to-Rail OTA for a common mode voltage of 0.5 V.

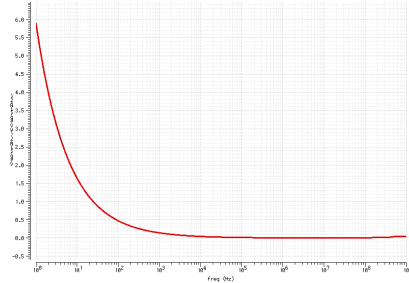


Fig. 20: Input Referred Noise for Common Mode Voltage = 0.5 V

Figure 21 shows the plot for the input-referred noise voltage on carrying out noise analysis on the Rail-to-Rail OTA for a common mode voltage of 1.3 V.

Table VII summarises the results obtained, such as maximum input referred noise (V_{in}) and integrated noise voltage (V_{int}), by performing noise analysis on the Rail-to-Rail OTA for various common mode voltages.

VIII. CONCLUSION

The Rail-to-Rail OTA with a constant transconductance value was designed and simulated in the Cadence Virtuoso software. It satisfies most of the requirements related to DC gain, unity gain bandwidth and phase margin within the common mode voltage range of 0.4 V to 1.3 V. However, it

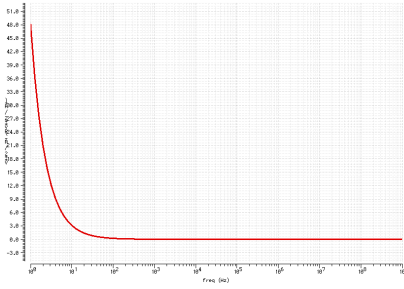


Fig. 21: Input Referred Noise for Common Mode Voltage = 1.3 V

V_{CM}	V_{in} (in $\mu V/\sqrt{Hz}$)	V_{int} (in mV)
0.4	8.6957	1.66757
0.5	5.8918	1.96705
0.6	5.643	0.759744
0.7	6.02384	2.25614
0.8	6.2415	9.364
0.9	6.3169	22.587
1	6.4107	10.485
1.1	6.4789	9.6
1.2	6.6303	9.532
1.3	6.9337	4.66

TABLE VII: Noise Analysis Results

does not satisfy the requirements of power consumption and CMRR. The table VIII shows the final summary of the design specification that are required and achieved.

Parameter	Requirement	Obtained
DC Gain	≥ 70 dB	70 - 100 dB
Unity Gain Bandwidth	≥ 40 MHz	37 - 154 MHz
Phase Margin	$60^\circ - 85^\circ$	67.4 - 98.63
Input Common Mode Range (ICMR)	200 mV to $(V_{supply} - 0.2)$ V	0.4 - 1.3 V
Output Common Mode Range (OCMR)	200 mV to $(V_{supply} - 0.2)$ V	0.4 - 1.3 V
Common Mode Rejection Ratio (CMRR)	≥ 90 dB	60 - 90 dB
Load Capacitance	≥ 5 pF	10 pF
Power consumption	≤ 1 mW	≥ 1 mW
Supply Voltage	≤ 1.8 V	≤ 1.8 V

TABLE VIII: Summary of Design Specifications

Note that there are variations in the design specifications achieved, with common mode voltage. The required DC gain, Unity Gain Bandwidth and Phase Margin are achieved for all voltage values within the ICMR range except for 0.6 V. The design specifications are not being met for 0.6 V due to improper biasing and operating regions of various transistors. Thus, there is a need to increase the common-mode voltage range from 0.2 to 1.6 V, reduce the power consumption significantly, ensure that the CMRR is satisfied and ensure that the circuit is robust and there isn't too much variation with common mode voltage. Other configurations for the input stage, such as square root current control and g_m control using two diodes, and the second stage, such as folded cascode, can be tried in order to satisfy all the specification requirements.

IX. REFERENCES

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